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Artificial Intelligence & Data Engineering Design Project Report

Title

JOINT KNOWLEDGE DISTILLATION AND TENSOR-BASED COMPRESSION OF DEEP NEURAL NETWORKS

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SUMMARY

The rapid growth of deep neural network complexity has created a critical gap between the accuracy achievable by state-of-the-art architectures and the computational resources available on edge devices such as autonomous drones, IoT sensors and mobile healthcare platforms. Existing compression approaches address this gap in isolation; tensor decomposition methods reduce structural size but suffer from gradient instability and catastrophic accuracy degradation when applied sequentially to pre-trained weights while knowledge distillation methods recover accuracy under compression but break down at high compression ratios due to the capacity gap and fail to handle the dimensional mismatches inherent in tensor-decomposed architectures. This project proposes a unified framework that resolves these limitations by integrating structural compression and knowledge distillation into a single joint optimization, constructing compressed student networks from scratch and training them under teacher supervision simultaneously rather than sequentially.

The framework combines three components. First, the PURSUhInT mechanism analyses the representational geometry of the teacher network using the mean squared Canonical Correlation Analysis coefficient (R^2_{CCA}) and applies k-means clustering to the layer representations to identify the most semantically distinct layers as distillation hint points, replacing arbitrary spatial heuristics with principled layer selection. Second, Empirical Variational Bayesian Matrix Factorization (EVBMF) is applied to the teacher's trained weight spectrum to derive a per-layer intrinsic rank map which is used to construct both a Canonical Polyadic (CP) decomposed and a Tucker decomposed student architecture before training begins, transferring the teacher's empirically observed capacity utilization into each student's structural blueprint. Third, both structurally heterogeneous students are co-trained simultaneously under the shared frozen teacher using spatial attention-transfer distillation at the PURSUhInT-selected hint points and output-space logit distillation with a dynamic loss weighting scheme that automatically balances the competing objectives without manual coefficient tuning.

The framework is evaluated on CIFAR-100 and ImageNet-100 across eight teacher–student architecture pairs spanning both homogeneous and cross-architecture settings. Results demonstrate substantial reductions in parameter count and computational cost while maintaining competitive accuracy relative to the uncompressed teacher baseline and outperforming standard knowledge distillation methods applied to identically compressed students.

1 INTRODUCTION

The rapid progress of deep learning in computer vision has been driven primarily by scale. Deeper and wider network architectures consistently deliver higher accuracy but at the cost of computational and memory footprints that have grown far faster than the hardware improvements available on commodity devices. Modern architectures such as ResNet, WideResNet and VGG routinely contain millions of parameters and require billions of floating-point operations per inference pass. Deploying these models on resource-constrained platforms such as autonomous drones, IoT sensors, handheld medical imaging devices or mobile smartphones is therefore infeasible without significant reduction of their computational cost. This bottleneck is widely described as the memory wall [1]: the cost of fetching a model's parameters from DRAM eclipses available compute bandwidth, rendering overparameterized models unsuitable for low-power deployment and making aggressive model compression an engineering imperative for practical edge AI.

1.1 The Limitations of Existing Approaches

Two families of techniques are conventionally applied to compress deep networks and both exhibit important limitations when used in isolation. Low-rank tensor factorization, specifically CP [2] and Tucker [3] decomposition, offers a mathematically principled route to structural compression by factorizing the four-dimensional weight tensors of convolutional layers into smaller components, reducing parameter count and FLOPs proportionally to the chosen rank. The standard pipeline, however, is sequential. It first decomposes a fully pre-trained network and then attempts to recover accuracy by fine-tuning under a shifted loss landscape. This process is inherently fragile; the decomposed layers exhibit vanishing gradients and catastrophic accuracy degradation at high compression ratios and requires multi-stage iterative fine-tuning to recover meaningful accuracy [4].

Knowledge Distillation (KD) [5] addresses accuracy recovery by training a compact student network under the supervision of a larger teacher, exploiting the teacher's softened output probabilities which encode richer inter-class relationships than one-hot labels to guide the student toward a better solution. However, logit-level distillation alone fails under extreme compression; the heavily compressed student lacks sufficient capacity to approximate the teacher's output distribution, a phenomenon known as the capacity gap [6]. Hint distillation mitigates this by aligning the student's intermediate feature maps with the teacher's [7] but existing feature-alignment methods require trainable 1×1 convolutional regressors to reconcile channel mismatches and depend on rigid spatial heuristics that often transfer redundant or noisy knowledge. These adapters add memory overhead during training and become structurally ill-defined when the student's intermediate layers have undergone tensor factorization. Since CP and Tucker decomposition produce non-standard layer sequences whose channel counts differ fundamentally from the teacher's. The PURSUhInT framework [8] improves the selection of which layers to align via representational clustering, yet it does not address structural compression itself.

1.2 The Proposed Framework

This project addresses the limitations above through a unified framework that integrates structural compression and knowledge distillation into a single joint training strategy, rather than applying them sequentially. The central hypothesis is that a student network constructed via tensor decomposition can be trained significantly more effectively when it is simultaneously guided by semantically meaningful supervision extracted from the teacher, calibrated to the student's compressed capacity and applied at layers that reflect distinct stages of the teacher's representational hierarchy.

The framework rests on three integrated components. The first is the PURSUhInT mechanism [8], which selects distillation targets by analysing the teacher's representational geometry rather than by position or heuristic. It computes the mean squared Canonical Correlation Analysis coefficient (R^2_{CCA}) [9] between the intermediate representations of all teacher layer pairs and applies k-means clustering to the layer representations using the pairwise $\mathrm{R}^2_{\text{CCA}}$ distance. The centroid layer of each cluster (the layer most representative of a semantically distinct processing stage) is selected as a hint point. The second component is an automatic rank selection strategy based on Empirical Variational Bayesian Matrix Factorization (EVBMF) [10] which estimates the intrinsic rank of each teacher layer's weight matrix from its singular value spectrum and transfers this per-layer rank map to the student before training begins. This replaces uniform global compression ratios with the teacher's empirically observed capacity utilization. The third component is a dual-student co-training architecture in which a CP-decomposed and a Tucker-decomposed student are constructed from scratch and trained simultaneously under the shared frozen teacher, supervised by a spatial attention-transfer distillation loss [11] applied at the PURSUhInT-selected hint points together with output-space logit distillation [5].

1.3 Contributions

The contributions of this project are as follows:

1. A joint optimization framework that simultaneously constructs and trains CP-decomposed and Tucker-decomposed student networks from scratch under a shared teacher, replacing the fragile sequential decompose-then-fine-tune pipeline [4].
2. Integration of the PURSUhInT algorithm [8] into the compressed-student training pipeline, demonstrating its effectiveness for selecting non-redundant, semantically informative hint points that guide structurally compressed, heterogeneous students.
3. Teacher-guided per-layer rank selection via EVBMF [10], providing automatic rank assignment calibrated to the teacher's observed information density without manual tuning or a held-out validation set.
4. Systematic evaluation across eight teacher–student architecture pairs on CIFAR-100 [12] and ImageNet-100 [13], reporting accuracy, parameter count, FLOPs and inference latency against multiple knowledge distillation baselines under identical compression configurations.

The remainder of this report is organized as follows. Section 2 surveys the relevant literature on knowledge distillation, tensor decomposition methods. Section 3 defines the system requirements, design constraints and engineering standards. Section 4 details the methodology. Section 5 presents experimental results and evaluation. Section 6 concludes with a summary of findings and directions for future work.

2 BACKGROUND

This section reviews the literature relevant to the two pillars of the proposed framework; knowledge distillation and low-rank tensor decomposition. It traces the limitations of each field in isolation and identifies the confluence of open problems that motivates a joint approach.

2.1 Knowledge Distillation

Knowledge distillation (KD) [5] transfers the functional behaviour of an overparameterized teacher network to a compact student network. Rather than training the student on hard one-hot labels alone, KD uses the teacher's softened output probability distributions which encode secondary relationships between classes as soft targets, exposing the student to richer supervisory signal than the ground truth provides. This concept was demonstrated by Hinton et al. [5] and has since become a foundational technique in model compression, offering a training-time alternative to post-hoc compression methods such as pruning [14] and quantization [15].

2.1.1 The Evolution of Logit-Based Distillation

Classical KD applies a temperature scalar $\mathcal{T} > 1$ to the softmax function before computing the cross-entropy between teacher and student output distributions. It spreads probability mass across non-target classes to expose the student to the teacher's dark knowledge. Subsequent work has extended this output-driven alignment in several directions. Spherical KD (SKD) [16] normalizes logits to preserve angular geometry between class representations. Decoupled KD (DKD) [17] separates target and non-target objectives, arguing that coupling them in a single cross-entropy term suppresses important correlations. Weighted Soft Labels (WSL) [18] dynamically calibrates the loss contribution of each training sample based on its difficulty while improving label efficiency in long-tail distributions.

2.1.2 The Capacity Gap

Despite these advances, logit-level distillation fails under extreme compression ratios. When a student is heavily compressed, its reduced parameterization is insufficient to reproduce the teacher's output distribution through logit matching alone. It’s a phenomenon known as the capacity gap [6]. At very high compression ratios, the student lacks the representational capacity to approximate the complex probability distributions generated by the teacher and increasing the distillation temperature or adjusting the loss weights provides only marginal relief. Logit distillation is therefore insufficient for the extremely low-rank tensor-decomposed students studied in this project, which require intermediate feature-level guidance to bridge the gap.

2.2 Feature and Attention-Based Distillation

2.2.1 Hint Distillation and Attention Transfer

To bypass the capacity gap, hint distillation forces the student to replicate the teacher's intermediate feature maps and converts the intractable global mapping problem into localized regression tasks. FitNets [7] introduced this idea by training the student to regress specific intermediate-layer outputs before global distillation begins. Attention Transfer (AT) [11] shifts this alignment from raw feature values to spatial attention maps, teaches the student where to allocate focus rather than strictly what features to extract. It defines the attention map of a feature tensor as the channel-averaged squared activation, normalized to unit Frobenius norm and minimizes the mean squared error between teacher and student attention maps. Variational Information Distillation (VID) [19] adopts an information-theoretic formulation; maximizing mutual information between teacher and student representations. Attention-based Feature Distillation (AFD) [20] further integrates dynamic search mechanisms to link intermediate hints adaptively.

2.2.2 Vulnerabilities of Feature Alignment Methods

These techniques share two structural vulnerabilities that become critical when the student architecture has been modified by tensor decomposition. First, mismatched channel dimensions between teacher and student require the insertion of trainable 1×1 convolutional regressors or projection matrices to reconcile feature map shapes. These adapters inflate memory overhead during training and introduce a secondary optimization objective that can interfere with the primary distillation signal. Second, all of these methods depend on rigid spatial heuristics for selecting which layers to align, typically matching layers by depth index or by the final layer of each spatial block. This assumes a fixed correspondence between teacher and student feature hierarchies that does not hold under heterogeneous compression, where the student's structural modification by CP [2] or Tucker [3] factorization disrupts the natural depth-to-depth correspondence.

2.2.3 Systematic Layer Selection via PURSUhInT

To replace flawed spatial heuristics, the PURSUhInT framework [8] quantifies semantic similarity between all teacher layers using Centered Kernel Alignment (CKA) [9] and the mean squared Canonical Correlation Analysis coefficient (R^2_{CCA}). CKA measures the similarity between two representation matrices in a way that is invariant to orthogonal transformations and isotropic scaling, providing a principled notion of representational distance. By applying k-means clustering to the L layer representations using the pairwise $\mathrm{R}^2_{\mathrm{CCA}}$ distance, PURSUhInT groups together teacher layers that carry redundant processing information and selects the mathematical centroid of each cluster as the optimal, non-redundant hint point. While PURSUhInT solves the layer selection problem, the broader class of feature alignment methods it informs remains constrained by trainable regressors and Euclidean feature losses that break down under the dimensional mismatches inherent in structurally heterogeneous tensor-decomposed architectures.

2.3 Tensor Decomposition for Model Compression

Low-rank tensor factorization provides a mathematically rigorous mechanism to structurally minimize the parameter space of a network prior to inference and offers a deterministic alternative to heuristic pruning.

2.3.1 Mechanics of CP and Tucker Decompositions

Canonical Polyadic (CP) decomposition [2] approximates a four-dimensional convolutional weight tensor $\mathcal{W}$ as a sum of $\mathcal{R}$ rank-one tensors. This factorization replaces the original convolution with a sequence of four lightweight operations; two 1×1 convolutions handling channel mixing and two depthwise convolutions applying separable spatial filtering along each axis independently. The total parameter count reduces from $C_{out} \times C_{in} \times k_H \times k_W$ to $\mathcal{R}(C_{in} + k_H + k_W + C_{out})$, yielding aggressive compression whenever $\mathcal{R}$ is much smaller than $\min(C_{in}, C_{out}) \frac{k_H \times k_W}{k_H + k_W}$. A consequence of the separable spatial decomposition is that CP strictly limits cross-channel interactions to the pointwise stages.

Tucker decomposition [3] factorizes $\mathcal{W}$ into a compact core tensor and two orthogonal factor matrices acting on the channel modes; $\mathcal{W} \approx \mathcal{G} \times_1 U_{out} \times_2 U_{in}$ where $\mathcal{G}$ is a small $\mathcal{R}_{out} \times \mathcal{R}_{in} \times k_H \times k_W$ core. This yields three sequential operations: a 1×1 convolution reducing channels, a full $k_H \times k_W$ convolution on the reduced channels, and a 1×1 convolution expanding channels. The full spatial kernel is preserved in the core convolution, allowing Tucker to maintain cross-channel correlations across all $k_H \times k_W$ positions, a capacity that CP's separable decomposition sacrifices. The independent rank pair $(\mathcal{R}_{out}, \mathcal{R}_{in})$ enables separate control over input and output channel compression and gives Tucker greater structural flexibility than a single global rank.

2.3.2 Structural Fragility and the Sequential Pipeline

Mathematical decomposition inherently disrupts pre-trained weight distributions. When fine-tuned, these networks frequently suffer from vanishing gradients and catastrophic accuracy degradation [4]. Existing methodologies typically deploy a single factorization technique uniformly across the network and attempt to stabilize it through multi-stage iterative fine-tuning. Deploying a single decomposition uniformly also ignores the complementary structural strengths of CP and Tucker. Without distillation, multi-stage fine-tuning remains fragile under extreme compression ratios [4].

2.4 Rank Estimation and Adaptive Compression

The efficacy of tensor factorization relies critically on determining the optimal rank for each layer. Selecting ranks that are too low discards representational capacity while selecting ranks that are too high preserves redundancy and sacrifices compression gains.

2.4.1 Variational Bayesian Matrix Factorization

Determining the optimal rank of a tensor is NP-hard in general [21]. A practical approach is Empirical Variational Bayesian Matrix Factorization (EVBMF) [10] which treats the weight matrix as a signal-plus-noise observation under a low-rank prior. EVBMF applies Bayesian shrinkage to the singular values of the weight matrix using the Marchenko–Pastur noise distribution as a threshold to retain only those singular values that exceed the noise floor. The resulting estimated intrinsic rank captures how much of a layer's total channel capacity is actively utilized in the teacher's learned representation. This method was operationalized in compression pipelines such as MUSCO [4] which estimate static ranks prior to factorization.

2.4.2 Limitations of Uniform and Static Rank Estimation

Applying EVBMF to the student's own randomly initialized weights or relying on uniform global rank ratios fails to leverage the teacher's richer, fully-trained weight spectrum. The resulting ranks reflect neither the teacher's actual per-layer capacity utilization nor the heterogeneous information density across layers. Early layers of deep networks typically maintain high intrinsic dimensionality to preserve spatial detail while intermediate layers exhibit lower intrinsic rank. It reflects the increasing abstraction of mid-level representations. A uniform rank ratio allocates too many parameters to redundant layers and too few to information-dense ones.

2.5 Summary

The reviewed literature reveals a confluence of open problems that motivate the present framework. Logit-level distillation collapses under extreme compression due to the capacity gap. Feature-alignment methods require trainable regressors that are incompatible with heterogeneous decomposed architectures. Single-topology decomposition ignores the complementary strengths of CP and Tucker factorization. Rank selection from student weights or uniform ratios misses the teacher's capacity signals. No existing framework co-trains structurally heterogeneous compressed students guided by non-redundant, semantically clustered hint points derived from the teacher's representational geometry. The framework proposed in this project addresses all of these limitations simultaneously, as detailed in Section 4.

3 SYSTEM REQUIREMENTS

3.1 Design Constraints and Relevant Engineering Standards

The development of the joint knowledge distillation and tensor-based compression framework is governed by several critical engineering constraints that define the boundaries of the design space. A primary constraint is computational complexity. Specifically, the integration of mean-squared CCA ($\mathrm{R}^2_{\mathrm{CCA}}$) for representational-similarity layer clustering and the execution of Canonical Polyadic (CP) or Tucker decomposition must be optimized to ensure that the overhead does not become prohibitive during the training phase. Furthermore, the system is bound by a minimum accuracy threshold, requiring the compressed student network to retain a significant percentage of the original teacher network's accuracy on benchmark datasets like CIFAR-100 and ImageNet-100 to remain viable for deployment. From a hardware perspective, the design is constrained by the latency and model size requirements of target edge platforms, such as autonomous vehicles and IoT devices, necessitating a substantial reduction in both FLOPs and total parameter counts. Finally, the project must adhere to project timelines.

In addition to these physical and operational constraints, the project complies with international engineering standards to ensure technical rigor and interoperability:

- ISO/IEC 23053:2022 – Framework for Artificial Intelligence (AI) Systems Using Machine Learning (ML): This standard provides a standardized vocabulary and architectural structure for this project. It ensures that descriptions of the Teacher-Student framework, knowledge transfer, and model components are technically consistent with international engineering norms.

- ISO/IEC TS 4213:2022 – Assessment of Machine Learning Classification Performance: This standard establishes the criteria for evaluating model performance. Sets the criteria for evaluating model performance on datasets. It ensures that metrics are measured and reported using a scientifically rigorous and comparable methodology.

3.2 Functional Requirements

1. The system shall utilize mean-squared CCA ($\mathrm{R}^2_{\mathrm{CCA}}$) to measure the pairwise representational similarity between all sub-block layers between all layers of the teacher network.

2. The system shall apply k-means clustering to the layer representations, using mean-squared CCA as the distance metric to group layers and identify distinct semantic stages which serve as informative hint points for guiding the student network.

3. The system shall factorize the convolutional layers of the student network using Canonical Polyadic (CP) and Tucker decomposition methods to reduce the total number of physical parameters and computational cost.

4. The system shall implement a composite training objective combining supervised cross-entropy loss, output-space logit distillation via KL divergence and spatial attention-transfer loss at the PURSUhInT-selected hint points, with automatic loss balancing via homoscedastic uncertainty weighting.

5. The system shall be capable of compressing and training standard deep learning architectures, specifically ResNet and VGG models.

6. The system shall process and evaluate model performance using the CIFAR-100 and ImageNet-100 datasets.

7. The system shall output measurable metrics including final accuracy, parameter reduction ratios and total FLOPs to verify the effectiveness of the compression.

3.3 Non-Functional Requirements

1. The compressed student model shall maintain a competitive level of accuracy, specifically incurring minimal loss compared to the original uncompressed teacher baseline on the CIFAR-100 and ImageNet-100 datasets.

2. The system shall achieve a significant reduction in model inference speed and latency, enabling the compressed network to function efficiently in real-time applications such as autonomous vehicle navigation.

3. The representational-similarity clustering and tensor decomposition processes must be computationally efficient enough to be integrated into the standard training phase without causing prohibitive overhead.

4. The final student model shall achieve a target compression ratio that significantly reduces the total parameter count, ensuring compatibility with the memory capacities of edge devices and IoT platforms.

5. The training framework shall maintain numerical stability and ensure gradient flow throughout the training of structurally compressed layers to prevent model divergence.

6. The system's performance evaluation and terminology shall strictly adhere to ISO/IEC 23053:2022 and ISO/IEC TS 4213:2022 to ensure scientific rigor and international interoperability.

3.4 Evaluation Methodology

The primary objective of this project is to develop a unified framework that achieves high-ratio model compression while preserving the predictive performance of state-of-the-art architectures. The following goals and evaluation criteria have been established to measure the success of the proposed methodology:

1. The project aims to achieve a substantial reduction in the physical parameter count for ResNet and VGG architectures. This will be evaluated by comparing the number of trainable parameters in the compressed student model against the original uncompressed teacher model.

2. The student model shall be trained to maintain competitive accuracy levels on the CIFAR100 and ImageNet-100 datasets without a drastic drop compared to the uncompressed teacher baseline. The success criterion is to stay within a specified percentage of the teacher's original Top-1 and Top-5 accuracy.

3. The project goals include demonstrating that hint points selected via the PURSUhInT approach are more effective for knowledge transfer than standard heuristic or random selection methods. This will be evaluated by comparing the convergence rate and final accuracy of student models trained with different hint selection strategies.

4. A measurable objective is to achieve a significant reduction in FLOPs. This will be evaluated using profiling tools to measure the computational cost of a single forward pass, ensuring the model is optimized for low-resource hardware.

5. Standardized Performance Assessment: All evaluations must comply with ISO/IEC TS 4213:2022 to ensure that metrics are measured and reported using a scientifically rigorous and comparable methodology.

4 SYSTEM ARCHITECTURE

This section describes the proposed framework in detail. Section 4.1 gives an overview of the five-stage pipeline. Sections 4.2–4.4 describe the student construction, rank selection and hint point selection components. Section 4.5 details the distillation objectives. Sections 4.6 and 4.7 describe the dual-student co-training architecture and training objective.

4.1 Framework Overview

The proposed framework jointly compresses a pre-trained teacher network T into two structurally heterogeneous student networks; a CP-decomposed student S_{CP} and a Tucker-decomposed student S_{Tuck} through a unified five-stage pipeline illustrated in Figure 1.

In the first stage, T is trained to convergence on the target dataset. In the second and third stages which run in parallel and offline before student training begins, two preparations are made. On the left branch (Figure 1), the PURSUhInT mechanism [8] extracts intermediate layer representations from T, computes pairwise representational distances, and applies k-means clustering to select K non-redundant hint points $H = \{h_1, \dots, h_K\}$ (the teacher layers that carry the most structurally distinct semantic information). On the right branch (Figure 1), EVBMF rank selection [10] analyses the teacher's trained weight spectrum to construct a per-layer rank map $\{\mathrm{r_i}\}$ that determines the structural blueprint of both students before training begins.

In the fourth stage, both students are constructed from scratch by decomposing every eligible convolutional layer via CP and Tucker factorization respectively using the rank map derived in Stage 3. The two students are co-trained simultaneously under the shared frozen teacher, supervised by an attention-transfer distillation loss applied at each PURSUhInT-selected hint point and output-space logit distillation. In the fifth stage, both compressed models are evaluated independently on accuracy, parameter count and FLOPs.

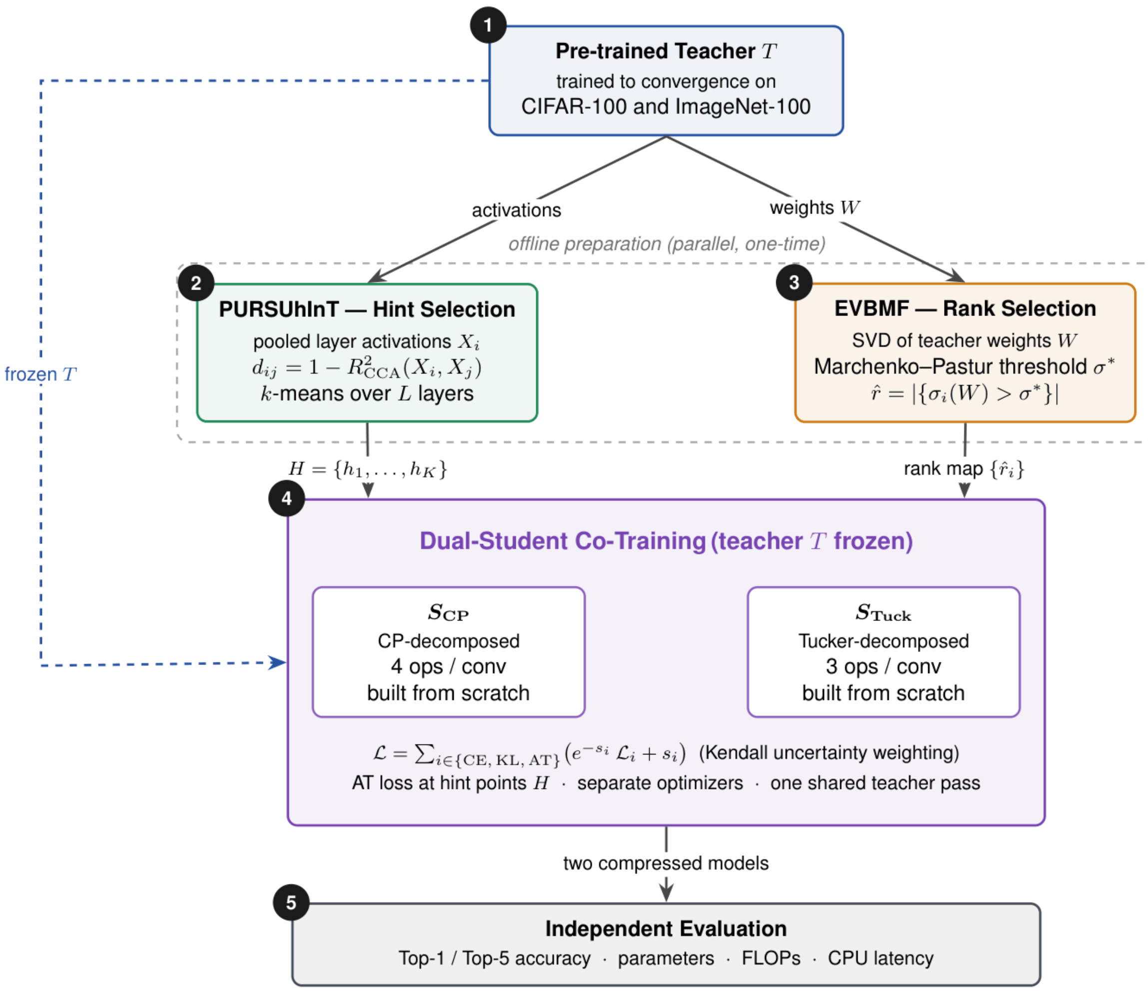

Figure 1: Overview of the proposed framework.

4.2 Student Architecture: Tensor Decomposition

Both student networks are constructed by replacing every eligible convolutional layer (defined as a $\mathrm{k_H} \times \mathrm{k_W}$ layer with $k_H > 1$ or $k_W > 1$) with a sequence of lightweight operations derived from low-rank tensor factorization. The factorized architecture is fixed before training begins. Only the layer structure is retained and all weight values are re-initialized from scratch. This distinguishes the approach from post-training compression where decomposition is applied to pre-trained weights that must then be fine-tuned under a shifted loss landscape.

4.2.1 Canonical Polyadic (CP) Decomposition

Canonical Polyadic (CP) decomposition [2] approximates a four-dimensional weight tensor $\mathcal{W} \in \mathbb{R}^{\mathrm{C_{out}} \times \mathrm{C_{in}} \times \mathrm{k_H} \times \mathrm{k_W}}$ as a sum of $\mathcal{R}$ rank-one tensors:

$$\mathcal{W} \approx \sum_{r=1}^{R} a_r \otimes b_r \otimes c_r \otimes d_r$$

where $\otimes$ denotes the outer product and R is the CP rank. This factorization replaces the original convolution with four sequential lightweight operations:

$$x \rightarrow (1 \times 1, C_{in} \rightarrow R) \rightarrow (k_H \times 1, \text{depthwise}) \rightarrow (1 \times k_W, \text{depthwise}) \rightarrow (1 \times 1, R \rightarrow C_{out}) \rightarrow y$$

The first and last 1×1 convolutions handle channel mixing; the two depthwise convolutions apply separable spatial filtering along each axis independently. The total parameter count reduces from $C_{out} \cdot C_{in} \cdot k_H \cdot k_W$ to $R(C_{in} + k_H + k_W + C_{out})$, yielding compression whenever $R \ll \min(C_{in}, C_{out}) \cdot \frac{k_H \cdot k_W}{k_H + k_W}$. A consequence of the separable spatial decomposition is that CP strictly limits cross-channel interactions to the pointwise stages.

4.2.2 Tucker Decomposition

Tucker decomposition [3] factorizes $\mathcal{W}$ into a compact core tensor and two orthogonal factor matrices acting on the channel modes:

$$\mathcal{W} \approx \mathcal{G} \times_1 U_{out} \times_2 U_{in}$$

where $\mathcal{G} \in \mathcal{R}^{R_{out} \times R_{in} \times k_H \times k_W}$ is the core, $U_{out} \in \mathcal{R}^{C_{out} \times R_{out}}$ and $U_{in} \in \mathcal{R}^{C_{in} \times R_{in}}$ are the factor matrices. This yields three sequential operations:

$$x \rightarrow (1 \times 1, C_{in} \rightarrow R_{in}) \rightarrow (k_H \times k_W, R_{in} \rightarrow R_{out}) \rightarrow (1 \times 1, R_{out} \rightarrow C_{out}) \rightarrow y$$

The full spatial kernel is preserved in the core convolution to allow Tucker to maintain cross-channel correlations across all $k_H \times k_W$ positions which is a capacity that CP's separable decomposition sacrifices. The independent rank pair (R_{out}, R_{in}) enables separate control over input and output channel compression, giving Tucker greater structural flexibility. The parameter count reduces from $C_{out} \cdot C_{in} \cdot k_H \cdot k_W$ to $R_{in} \cdot C_{in} + R_{in} \cdot R_{out} \cdot k_H \cdot k_W + R_{out} \cdot C_{out}$.

4.3 Teacher-Guided Rank Selection via EVBMF

Selecting the per-layer decomposition rank is critical. Ranks that are too low discard representational capacity while uniform global rank ratios (setting every layer's rank to a fixed fraction of its channel count) ignore the heterogeneous information density across layers. The framework instead derives per-layer ranks from the teacher's fully-trained weight spectrum using EVBMF [10].

EVBMF treats the weight matrix as a signal-plus-noise observation under a low-rank prior. Applied to the mode-0 unfolding $\mathcal{W} \in \mathcal{R}^{C_{out} \times (C_{in} \cdot k_H \cdot k_W)}$ of a convolutional layer, EVBMF applies Bayesian shrinkage to the singular values using the Marchenko–Pastur noise distribution as a threshold. It retains only those singular values that exceed the noise floor:

$$\hat{R} = |\sigma_i(\mathcal{W}) : \sigma_i(\mathcal{W}) > \sigma^*(\mathcal{W})|$$

where $\sigma^*(\mathcal{W})$ is the empirical noise threshold and $\hat{R}$ is the estimated intrinsic rank. It captures how much of the layer's total channel capacity is actively utilized.

Because the student and teacher typically differ in depth and width, student layer i is matched to teacher layer j via positional interpolation over the set of decomposable convolutions:

$$j = round\left(\frac{i \cdot (T-1)}{S-1}\right)$$

where S and T are the total numbers of decomposable convolutions in the student and teacher respectively. The teacher's rank is then expressed as a utilization fraction of its output channel count and scaled to the student's channel dimensions. For CP decomposition:

$$\mathrm{R}_{\mathrm{i}}^{(\mathrm{CP})} = \mathrm{round}\left(\frac{\hat{R}_{\mathrm{j}}}{\mathrm{C}_{\mathrm{out,j}}^{(\mathrm{T})}} \cdot \max\left(\mathrm{C}_{\mathrm{out,i}}^{(\mathrm{S})}, \mathrm{C}_{\mathrm{in,i}}^{(\mathrm{S})}\right)\right)$$

For Tucker decomposition, EVBMF is applied independently to the mode-0 unfolding (output channels) and the mode-1 unfolding (input channels), yielding a separate rank pair $(R_{out,i}, R_{in,i})$ for each student layer. The resulting rank map is computed once before training begins and used to construct the decomposed student architectures. Layer dimensions are fixed for the entire training run.

4.4 Non-Redundant Hint Point Selection via PURSUhInT

Rather than selecting distillation targets by hand or applying rigid spatial heuristics, the framework adopts the PURSUhInT mechanism [8] to identify K non-redundant hint points from the teacher's layer hierarchy (left branch, Figure 1).

For each of the teacher's L sub-blocks, a forward pass over a random subset of the training set is performed with spatial average pooling to obtain an $N \times C_i$ representation matrix X_i. Pairwise representational distances between all layers are computed as:

$$\mathrm{d}(\mathrm{i},\mathrm{j}) = 1 - \mathrm{R}_{\mathrm{CCA}}^2\left(\mathrm{X}_{\mathrm{i}}, \mathrm{X}_{\mathrm{j}}\right)$$

where R_{CCA}^2 [9] is the mean squared Canonical Correlation Analysis coefficient, measuring the proportion of variance explained by the optimal linear mapping between the two representations. K-means clustering is applied to the L layer representations using R_{CCA}^2 as the distance function, grouping layers that carry redundant semantic information into K clusters. The index closest to the mean index of each cluster is selected as the centroid, yielding the final hint point set $\mathcal{H} = \{h_1, \dots, h_K\} \subset \{1, \dots, L\}$. These K indices identify the teacher layers at which distillation supervision is applied during student training.

4.5 Distillation Objectives

The framework applies distillation supervision at each of the K PURSUhInT-selected hint points. Two complementary objectives are used; output-space logit distillation and spatial attention transfer.

4.5.1 Output-Space Knowledge Distillation

Output-space distillation [5] supervises the student's final classification layer using a temperature-scaled Kullback–Leibler divergence between the teacher's and student's softened probability distributions:

$$L_{KL} = KL\left(\sigma\left(\frac{z_t}{\tau_{KD}}\right) \parallel \sigma\left(\frac{z_s}{\tau_{KD}}\right)\right)$$

where z_t and z_s are the teacher's and student's logit vectors, σ is the softmax function and $\tau_{KD} = 4$ is the distillation temperature. This provides coarse output-level guidance and acts as a regularizer once the students develop structured intermediate representations.

4.5.2 Spatial Attention Transfer

Following Zagoruyko and Komodakis [11], the spatial attention map of a feature tensor is defined by squaring all activations, averaging across the channel dimension and ℓ_2-normalizing the result. For a batch of B samples at hint point h_k with student feature tensor $f_s^k \in \mathcal{R}^{B\times C_s\times H_s\times W_s}$ and teacher feature tensor $f_t^k \in \mathcal{R}^{B\times C_t\times H_t\times W_t}$:

$$v^k = \left(\frac{1}{C}\right)\sum_c \left(f^k_{:,c,:,:}\right)^2 \in \mathbb{R}^{B\times H_t\times W_t}$$

$$a^k = \frac{v^k}{\left||v^k|\right|_2}$$

When the student and teacher feature maps differ in spatial resolution, adaptive average pooling is applied to both the student and teacher feature maps so that they are compared at their common minimum spatial resolution; $f_s^k = \mathrm{AvgPool}\left(f_s^k, (H_t, W_t)\right)$. The attention matching loss at hint point k is then:

$$\mathcal{L}_{AT}^k = \mathrm{MSE}\left(a_s^k, a_t^k\right)$$

where $\mathrm{MSE}(A, B) = \left(\frac{1}{|A|}\right)\left||A - B|\right|_F^2$ is the mean squared error over all elements. This term is sign-invariant by construction and requires no trainable regressor, since the spatial dimensions are reconciled entirely by pooling. The total attention transfer loss sums over all K hint points:

$$\mathcal{L}_{\mathrm{AT}} = \sum_{k=1}^{K} \mathcal{L}_{\mathrm{AT}}^{k}$$

4.6 Heterogeneous Dual-Student Co-Training

Both student networks are derived from the same base architecture (e.g., ResNet-18) but decomposed by different methods: S_{CP} via CP decomposition and S_{Tuck} via Tucker decomposition, each using the per-layer rank map derived in Section 4.3. The two students are structurally heterogeneous (their layer configurations, intermediate channel counts and parameter counts differ) but their depth and the positions of their hint-point connectors align with the student hint indices matched to the student's block boundaries.

Both students are initialized from scratch and trained simultaneously within a single training loop. They share the same data loader and teacher forward pass at each step to reduce co-training overhead to roughly twice the cost of single-student distillation. Each student maintains entirely separate optimizers, learning rate schedulers, gradient scalers and checkpoint trackers. Neither student shares gradients with the other.

The motivation for simultaneous training is efficiency. The two students share a single teacher forward pass per training step to avoid the three-pass cost of sequential training.

4.7 Training Objective

Each student is trained with a composite loss combining supervised classification and output-space distillation with the attention-transfer objective. For a student $S \in \{S_{CP}, S_{Tuck}\}$ the full per-step objective is:

$$\mathcal{L}_{\mathrm{S}} = \gamma\mathcal{L}_{\mathrm{CE}} + \alpha\mathcal{L}_{\mathrm{KL}} + \beta\mathcal{L}_{\mathrm{AT}}$$

Where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss with ground-truth labels, $\mathcal{L}_{\mathrm{KL}}$ is the Kullback–Leibler divergence between the student's and teacher's softened logits at temperature $\tau_{KD} = 4$, $\mathcal{L}_{\mathrm{AT}} = \sum_{k} \mathcal{L}_{\mathrm{AT}}^{k}$ is the total attention transfer loss summed across K hint points, and γ, α, β are scalar loss weights.

As an alternative to fixed scalar weights, the framework supports the homoscedastic uncertainty weighting of Kendall et al. [22] which replaces fixed coefficients with learnable log-variance parameters $\{s_i\}$:

$$\mathcal{L}_{\mathrm{S}}^{\mathrm{dyn}} = \sum_{i \in \{\mathrm{CE,KL,AT}\}} \left[e^{-s_i}\,\mathcal{L}_i + s_i\right]$$

Each s_i is initialized to zero and placed in a separate parameter group with no weight decay to prevent divergence. The uncertainty weighting automatically balances tasks with different loss scales throughout training and eliminate the need for manual coefficient tuning.

The hyperparameters used in all experiments are; $\tau_{KD} = 4, K = 3$ hint points for CIFAR and $K = 4$ for ImageNet-100, dynamic loss weighting enabled.

5 RESULTS AND EVALUATION

5.1 Experimental Setup

The framework is evaluated on two image-classification benchmarks. CIFAR-100 [12] consists of $60{,}000$ colour images of resolution 32×32, evenly distributed across 100 classes and split into $50{,}000$ training and $10{,}000$ test images. ImageNet-100 is a 100-class subset of the ILSVRC-2012 dataset [13]; validation images are resized to 256 pixels on the shorter side and centre-cropped to 224×224. CIFAR-100 serves as a low-resolution stress test in which extreme compression ratios become achievable whereas ImageNet-100 evaluates the framework at full input resolution and greater task difficulty.

Eight teacher-student configurations are evaluated: two on ImageNet-100 and six on CIFAR-100. The ImageNet-100 pairs are ResNet-34 → ResNet-18 and ResNet-50 → ResNet-18. The six CIFAR-100 pairs are WRN-40-2 → WRN-16-2, WRN-40-2 → ResNet-8×4, ResNet-32×4 → ResNet-8×4, ResNet-32×4 → WRN-16-2, ResNet-110 → ResNet-20 and VGG-13 → VGG-8. Two of the CIFAR-100 pairs (WRN-40-2 → ResNet-8×4 and ResNet-32×4 → WRN-16-2) are cross-architecture, in which teacher and student belong to different network families; the remaining pairs are homogeneous. This selection spans residual, wide-residual and plain-convolutional topologies and includes both same-family and cross-family knowledge transfer.

All students are trained from scratch using stochastic gradient descent with momentum 0.9. On CIFAR-100, training runs for 240 epochs with a batch size of 64, an initial learning rate of 0.05 decayed by a factor of ten at epochs $150, 180, 210$, and a weight decay of 5×10^{-4}. On ImageNet-100, training runs for 100 epochs with a batch size of 256, an initial learning rate of 0.1 (scaled linearly with the batch size) decayed by a factor of ten at epochs $30, 60$, and 90, and a weight decay of 1×10^{-4}. Both schedules apply a five-epoch linear learning-rate warmup. PURSUhInT selects $\mathrm{K} = 3$ hint points on CIFAR-100 and $\mathrm{K} = 4$ on ImageNet-100, using the mean-squared CCA distance $(1 - \mathrm{R}^2_{\mathrm{CCA}})$. The distillation temperature is $\tau = 4$. Per-layer ranks are obtained by applying EVBMF to the teacher's trained weight spectrum and projecting them onto each student through positional interpolation (Section 4.3). The cross-entropy, KL-divergence, and attention-transfer terms are balanced automatically by the homoscedastic uncertainty weighting of Section 4.7, with the log-variance parameters placed in a separate parameter group exempt from weight decay. The CP and Tucker students are co-trained simultaneously under a single shared teacher forward pass, each with its own optimizer, scheduler and gradient scaler.

Because the contribution of this project is the compression framework rather than any single distillation loss, every configuration is evaluated under five distillation objectives applied within the identical pipeline (the same EVBMF-derived ranks, the same PURSUhInT hint points and the same dual-student co-training loop) with only the distillation criterion changed. The five objectives are logit-only Knowledge Distillation (KD) [5], FitNet hint regression [7], Attention Transfer (AT) [11], Variational Information Distillation (VID) [19] and Weighted Soft Labels (WSL) [18]. KD applies no intermediate-feature supervision and therefore serves as the logit-only reference while the framework adopts Attention Transfer as its default objective in light of its consistently strong performance across configurations (Sections 5.2–5.3).

Each compressed student is assessed on four axes; Top-1 and Top-5 classification accuracy, total parameter count, floating-point operations (FLOPs) per forward pass and single-image inference latency. Parameters and FLOPs are measured on the deployed model with no training-time adapter overhead using the THOP profiler. Compression ratios are reported relative to the corresponding uncompressed teacher. All accuracy measurement and reporting follow ISO/IEC TS 4213:2022 to ensure a scientifically rigorous and comparable methodology.

5.2 ImageNet-100 Results

Table 1: Results on ImageNet-100. Compression ratios are relative to the respective teacher; the best Top-1 per column is shown in bold.

Method	Decomp.	Top-1 (%)	Top-5 (%)	Params	FLOPs	Params Compr.	FLOPs Compr.
ResNet-34 → ResNet-18							
Teacher (ResNet-34)	—	81.72	95.38	$21.34\,M$	$3.68\,G$	$1.0\times$	$1.0\times$
KD [5]	CP	77.16	93.68	$1.33\,M$	$277.99\,M$	$16.04\times$	$13.23\times$
FitNet [7]		78.48	94.70				
AT [11]		**79.50**	**95.00**				
VID [19]		78.52	94.54				
WSL [18]		77.44	94.42				
KD [5]	Tucker	77.26	94.14	$3.25\,M$	$655.50\,M$	$6.56\times$	$5.60\times$
FitNet [7]		**80.22**	**95.34**				
AT [11]		80.12	95.32				
VID [19]		79.88	95.14				
WSL [18]		79.44	95.16				
ResNet-50 → ResNet-18							
Teacher (ResNet-50)	—	82.54	95.96	$23.71\,M$	$4.13\,G$	$1.0\times$	$1.0\times$
KD [5]	CP	78.10	93.94	$1.35\,M$	$270.15\,M$	$17.59\times$	$15.29\times$
FitNet [7]		78.48	94.44				
AT [11]		**79.60**	**94.84**				
VID [19]		78.14	94.04				
WSL [18]		78.58	94.52				
KD [5]	Tucker	77.66	93.94	$3.18\,M$	$600.63\,M$	$7.46\times$	$6.88\times$
FitNet [7]		80.10	95.28				
AT [11]		**81.04**	**95.56**				
VID [19]		80.18	95.48				
WSL [18]		80.24	95.34				

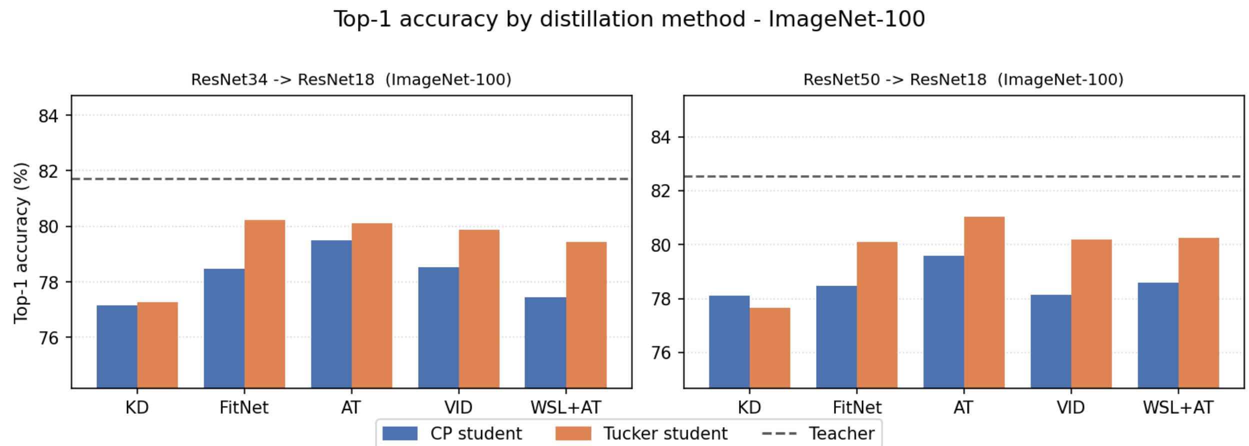

Figure 2: Top-1 accuracy of CP and Tucker students under five distillation objectives on ImageNet-100 (ResNet-34→18 and ResNet-50→18).

Table 1 presents Top-1 and Top-5 accuracy alongside compression metrics for two teacher-student configurations on ImageNet-100. In both configurations the student is ResNet-18, trained under the full pipeline: PURSUhInT hint-point selection ($K = 4$ clusters), EVBMF rank estimation, CP and Tucker decomposition trained in parallel and Kendall uncertainty-weighted loss combining cross-entropy, knowledge distillation and attention transfer [11]. Figure 2 provides a visual summary of the method ranking under both decompositions for both configurations.

For the ResNet-34 → ResNet-18 configuration, the CP-decomposed student achieves 79.50% Top-1 accuracy within 2.22 percentage points (pp) of the teacher (81.72%) while compressing parameters by 16.04× (1.33 M vs. 21.34 M) and FLOPs by 13.23×. The Tucker-decomposed student reaches 80.12%, narrowing the accuracy gap to 1.60 pp at a more moderate 6.56× parameter compression. Across all five distillation objectives evaluated, feature-based methods consistently outperform logit-only KD on both decompositions. AT achieves the highest accuracy among the CP variants (79.50%) and ranks second among Tucker variants (80.12%) where FitNet achieves the column best of 80.22%, a margin of 0.10 pp that lies within expected run-to-run variability. The advantage of feature-based methods over KD is most pronounced on Tucker ($+2.86$ pp for AT vs. KD), confirming that intermediate-layer supervision provides a richer training signal than output-level distillation alone.

The ResNet-50 → ResNet-18 configuration is structurally more demanding: the teacher is deeper and wider than in the previous configuration while the student architecture is identical. Here, CP achieves 79.60% Top-1 at 17.59× parameter compression and Tucker achieves 81.04% at 7.46×, only 1.50 pp below the ResNet-50 teacher (82.54%). AT is the best-performing method on both CP and Tucker for this configuration. The Tucker accuracy advantage over CP widens to 1.44 pp (versus 0.62 pp in the ResNet-34 configuration). It suggests that Tucker's two-stage decomposition (which retains the full $k \times k$ convolution core and preserves cross-channel correlations) becomes increasingly beneficial as the compression ratio grows and the capacity gap between teacher and student widens.

Taken together, the ImageNet-100 results show that the framework achieves substantial compression on a 224×224-resolution benchmark with only a modest accuracy drop. Tucker decomposition is the preferred choice when the target compression ratio is moderate (≈ 6–8×) and top accuracy is the priority. CP decomposition is preferable when the tightest possible parameter budget is required, accepting a slightly larger accuracy penalty in return for a factor-of-two additional compression over Tucker. In both cases, CPU inference latency drops from 265.98 ms (teacher) to 135.88 ms with CP (1.96×) and 125.91 ms with Tucker (2.11×). It confirms real-world deployment benefits beyond parameter counts alone. Figure 3 plots these results as an accuracy–compression Pareto frontier for the ResNet-34 → ResNet-18 configuration.

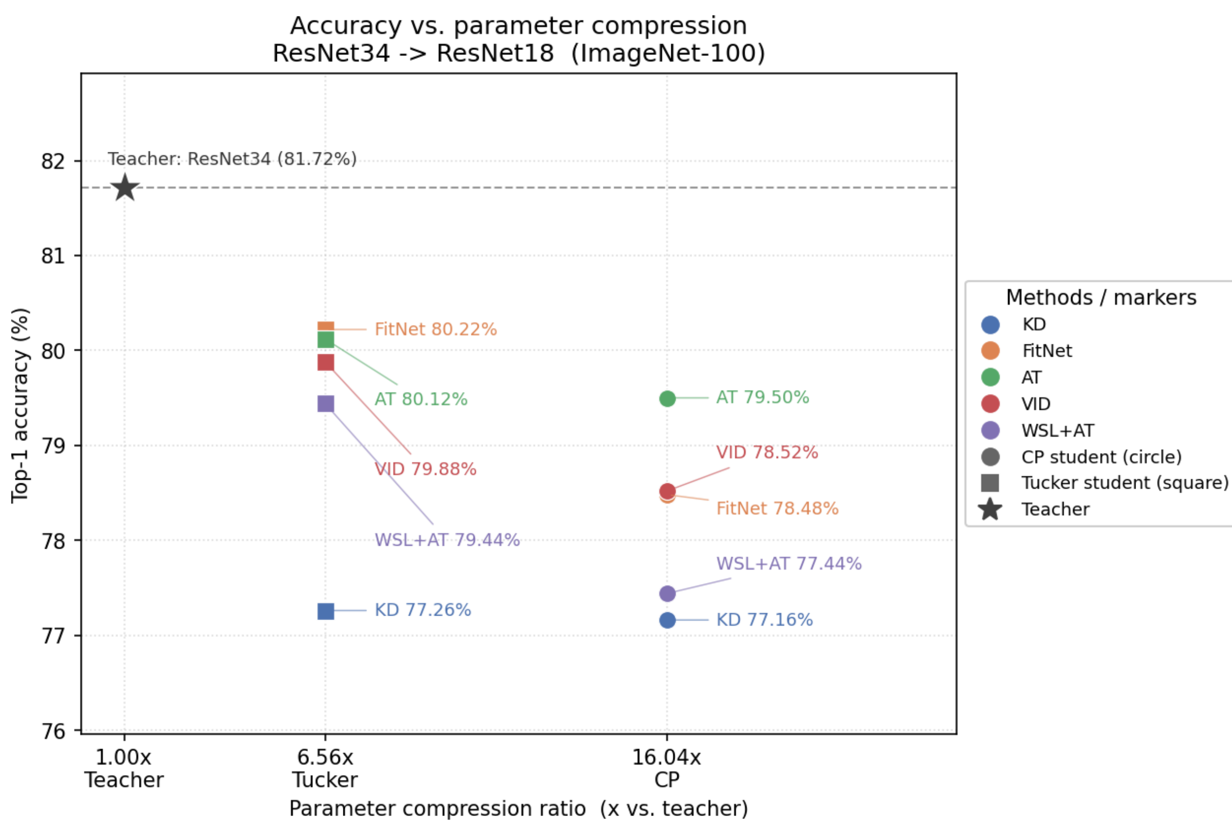

Figure 3: Accuracy–compression Pareto plot for the ResNet-34→ResNet-18 configuration. Each point is one (method, decomposition) pair.

5.3 CIFAR-100 Results

Table 2: Results on CIFAR-100 (configurations 1-3 of 6). Best Top-1 per column in bold.

Method	Decomp.	Top-1 (%)	Top-5 (%)	Params	FLOPs	Params Compr.	FLOPs Compr.
WRN-40-2 → WRN-16-2							

Teacher (WRN-40-2)	—	76.79	93.60	2.26 M	330.43 M	1.0 ×	1.0 ×
KD [5]	CP	69.01	91.63	111.80 K	17.20 M	20.27 ×	19.21 ×
FitNet [7]		71.47	92.80				
AT [11]		**71.85**	**93.00**				
VID [19]		71.34	92.47				
WSL [18]		70.59	92.79				
KD [5]	Tucker	68.79	91.17	332.76 K	52.03 M	6.78 ×	6.35 ×
FitNet [7]		71.95	92.97				
AT [11]		**72.23**	**93.16**				
VID [19]		71.33	92.75				
WSL [18]		71.82	92.87				
WRN-40-2 → ResNet-8×4 (cross-architecture)							
Teacher (WRN-40-2)	—	76.01	93.79	2.26 M	330.43 M	1.0 ×	1.0 ×
KD [5]	CP	73.27	93.45	258.77 K	39.05 M	8.71 ×	8.46 ×
FitNet [7]		72.65	93.11				
AT [11]		**73.56**	**93.50**				
VID [19]		72.29	93.08				
WSL [18]		72.30	93.11				
KD [5]	Tucker	71.61	93.13	726.67 K	92.24 M	3.10 ×	3.58 ×
FitNet [7]		72.03	92.82				
AT [11]		72.20	93.28				
VID [19]		**72.25**	**92.98**				
WSL [18]		71.22	92.81				
ResNet-32×4 → ResNet-8×4							
Teacher (ResNet-32×4)	—	79.61	94.94	7.43 M	1.09 G	1.0 ×	1.0 ×
KD [5]	CP	71.88	92.40	252.53 K	48.77 M	29.44 ×	22.31 ×
FitNet [7]		72.44	92.78				
AT [11]		**73.54**	**93.49**				
VID [19]		72.79	92.92				
WSL [18]		73.08	93.37				
KD [5]	Tucker	70.85	92.59	679.70 K	130.57 M	10.94 ×	8.33 ×
FitNet [7]		71.70	92.98				
AT [11]		**73.14**	**93.63**				
VID [19]		71.44	92.54				
WSL [18]		72.17	93.22				

Table 3: Results on CIFAR-100 (configurations 4–6 of 6). Best Top-1 per column in bold.

Method	Decomp.	Top-1 (%)	Top-5 (%)	Params	FLOPs	Params Compr.	FLOPs Compr.
ResNet-32×4 → WRN-16-2 (cross-architecture)							
Teacher (ResNet-32×4)	—	79.64	94.69	7.43 M	1.09 G	1.0 ×	1.0 ×
KD [5]	CP	66.02	89.78	116.09 K	20.83 M	64.04 ×	52.24 ×
FitNet [7]		71.36	92.50				
AT [11]		**72.01**	**93.26**				
VID [19]		69.29	91.46				
WSL [18]		71.47	92.71				
KD [5]	Tucker	68.25	91.23	340.19 K	70.65 M	21.85 ×	15.40 ×
FitNet [7]		71.59	92.92				
AT [11]		**73.06**	**93.46**				
VID [19]		68.52	91.58				
WSL [18]		72.05	93.06				
ResNet-110 → ResNet-20							
Teacher (ResNet-110)	—	74.30	93.21	1.74 M	257.40 M	1.0 ×	1.0 ×
KD [5]	CP	63.64	89.00	47.60 K	8.35 M	36.48 ×	30.83 ×
FitNet [7]		66.67	90.60				
AT [11]		66.06	90.37				
VID [19]		**66.74**	**91.12**				
WSL [18]		65.23	90.25				
KD [5]	Tucker	65.79	90.61	127.24 K	23.00 M	13.65 ×	11.19 ×
FitNet [7]		**67.43**	**91.35**				
AT [11]		67.15	90.92				
VID [19]		67.09	91.33				
WSL [18]		66.72	90.72				
VGG-13 → VGG-8							
Teacher (VGG-13)	—	75.21	92.49	9.46 M	285.99 M	1.0 ×	1.0 ×
KD [5]	CP	**70.79**	**91.44**	574.08 K	14.79 M	16.48 ×	19.34 ×
FitNet [7]		70.20	91.34				
AT [11]		70.26	91.49				
VID [19]		70.28	91.35				
WSL [18]		69.10	90.97				
KD [5]	Tucker	**69.85**	**91.29**	1.63 M	40.21 M	5.82 ×	7.11 ×
FitNet [7]		68.94	90.60				
AT [11]		69.22	91.03				
VID [19]		68.55	90.86				

WSL [18]		67.98	90.36				

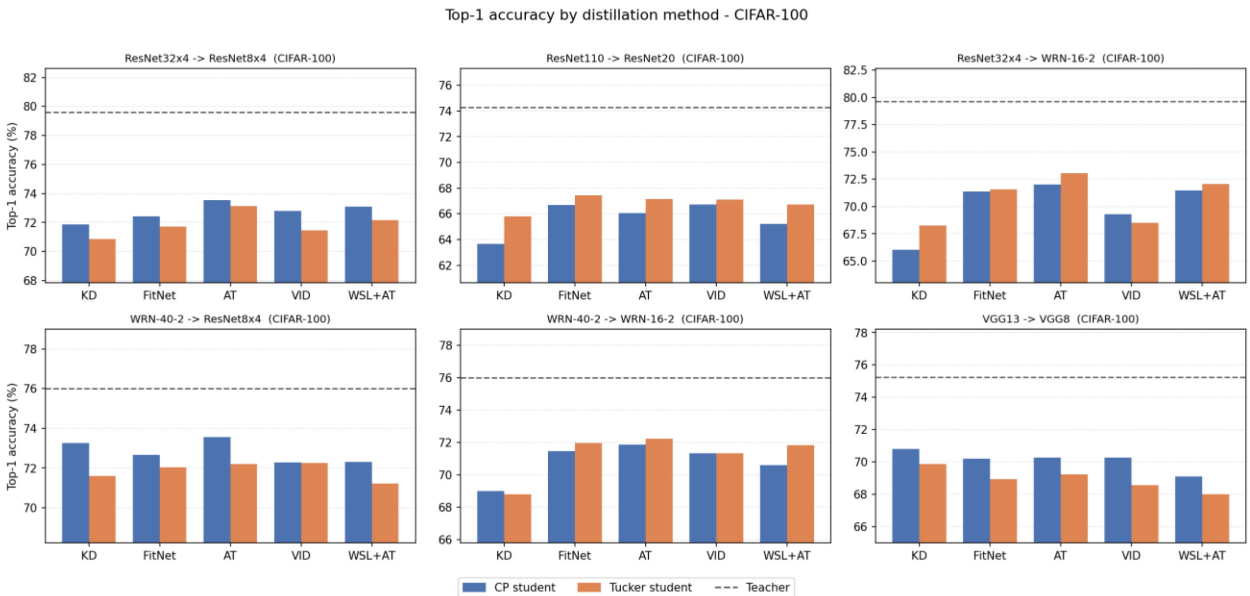

Figure 4: Top-1 accuracy of CP and Tucker students under five distillation objectives on CIFAR-100

Tables 2 and 3 report results for six teacher-student configurations on CIFAR-100. The same pipeline used for ImageNet-100 is applied unchanged, with $K = 3$ PURSUhInT clusters and 240 training epochs. The configurations span same-architecture and cross-architecture pairs and cover a wide range of compression ratios. Figure 4 provides a visual overview of the method ranking across all six CIFAR-100 configurations.

WRN-40-2 → WRN-16-2 compresses parameters by 20.27× (CP) and 6.78× (Tucker). AT achieves 71.85% Top-1 with CP and 72.23% with Tucker against a teacher accuracy of 76.79% with feature-based methods uniformly outperforming logit-only KD by up to $\sim$3.4 pp. ResNet-32×4 → ResNet-8×4 produces a 29.44× CP compression ratio and yields AT-CP 73.54%, AT-Tucker 73.14% against the teacher's 79.61%. ResNet-110 → ResNet-20 reaches the most aggressive same-architecture compression: 36.48× with CP (47.60 K parameters, 8.35 M FLOPs) and 13.65× with Tucker. At this compression level the accuracy drop is larger (AT-CP 66.06% and AT-Tucker 67.15% versus a teacher of 74.30%) and the best results shift; VID leads on CP at 66.74% and FitNet leads on Tucker at 67.43%, both by a margin of less than 0.70 pp over AT. This suggests that at very high compression ratios the relative ranking of distillation objectives becomes less stable. It’s likely because the student's limited capacity constrains what any supervision signal can achieve.

The framework handles teacher-student pairs with different architectures without modification. The AT loss computes spatial attention maps independently for each network and the EVBMF rank estimator maps teacher layers to student layers by relative position. WRN-40-2 → ResNet-8×4 achieves AT-CP 73.56% (8.71× params) and AT-Tucker 72.20% (3.10× params, where VID matches AT to within 0.05 pp). ResNet-32×4 → WRN-16-2 produces the largest compression ratio in this study; 64.04× with CP (116.09 K parameters) against a teacher accuracy of 79.64%. Despite this extreme compression, AT-CP achieves 72.01% and AT-Tucker achieves 73.06% at 21.85× compression. It is a result that demonstrates the benefit of structured intermediate-layer supervision from a high-capacity teacher for guiding a structurally different and very compact student.

The VGG-13 → VGG-8 pair is the only configuration in this study where logit-only KD outperforms all feature-based methods; KD achieves 70.79% with CP and 69.85% with Tucker while AT yields 70.26% and 69.22% respectively. VGG architectures differ from ResNets and WRNs in that they lack residual connections and rely on a large fully connected head. Their internal spatial representations may be less structured and make attention-map transfer less effective. The framework's EVBMF rank estimator assigns relatively higher ranks to VGG's wide convolutional layers (CP: $574\ K$ parameters at 16.48× compression). It indicates that capacity is not the limiting factor. This result underscores that the choice of distillation objective remains architecture-dependent; AT [11], while highly effective across most configurations (7 of 12 CIFAR columns), it's not universally dominant.

Across all 12 evaluated columns (six configurations $\times$ two decompositions), AT is the best-performing distillation loss in 7 cases. The five remaining exceptions are the VGG pair (both columns where KD wins), WRN-40-2 → ResNet-8×4 Tucker (VID wins by 0.05 pp), ResNet-110 → ResNet-20 CP (VID wins by 0.68 pp) and ResNet-110 → ResNet-20 Tucker (FitNet wins by 0.28 pp). These results confirm that combining tensor decomposition with structured feature distillation enables aggressive compression across diverse architectures; the framework achieves up to 64× parameter reduction while maintaining accuracy within 7–8 pp of the teacher on the hardest configurations and within 3–5 pp on moderate-compression configurations without any architecture-specific tuning.

5.4 Compression–Accuracy Trade-off and Efficiency

The eight configurations in Tables 1–3 trace a consistent compression-accuracy trade-off illustrated by the bar charts of Figures 2 and 4 and the Pareto frontier of Figure 3. In every configuration the CP-decomposed student attains the higher compression ratio while the Tucker-decomposed student attains the higher accuracy and this ordering never reverses. The separation is structural; CP factorises each convolution into four separable operations whose parameter cost scales with a single rank and discards all cross-channel interaction outside the pointwise stages whereas Tucker retains a dense $k \times k$ core convolution that preserves cross-channel correlations at the cost of a larger parameter budget (Section 4.2). For the ResNet-34 → ResNet-18 pair, this yields $1.33\ M$ parameters at 16.04× compression for CP against $3.25\ M$ parameters at 6.56× for Tucker — a 2.4× difference in parameter count driven entirely by the structural choice of decomposition.

The magnitude of the trade-off scales with input resolution. On 32×32 CIFAR-100 inputs, EVBMF identifies very low intrinsic ranks and drives CP compression ratios of 20–64× across the six configurations. The most extreme case, ResNet-32×4 → WRN-16-2 reaches 64.04× with CP ($116\ K$ parameters) while still achieving 72.01% Top-1 under AT supervision. This result demonstrates the benefit of strong feature-level guidance at very low capacities. On 224×224 ImageNet-100 inputs the same estimator yields more conservative ratios of 16–18× (CP) and 6–8× (Tucker), reflecting the higher intrinsic dimensionality required to process full-resolution spatial features. In both regimes the accuracy retained relative to the teacher is high; within 1.6 pp on ImageNet-100 with Tucker and within 3–5 pp on the moderate-compression CIFAR-100 pairs.

These compression ratios translate directly into measured deployment speedups. For the ResNet-34 → ResNet-18 pair, single-image CPU inference latency falls from 265.98 ms (teacher) to 135.88 ms with CP (1.96×) and 125.91 ms with Tucker (2.11×). Notably, despite the Tucker student carrying 2.4× more parameters than CP, it achieves the lower latency of the two. Its three-operation factorisation (1×1 → $k \times k$ core → 1×1) issues fewer sequential kernel launches than CP's four-operation chain (1×1 → depthwise-H → depthwise-W → 1×1) and its dense core convolution maps more efficiently than CP's depthwise separable stages. Parameter count alone therefore understates Tucker's deployment advantage and confirms it as the preferred choice whenever moderate compression is acceptable.

5.5 Per-Layer Rank Allocation

Figure 5 shows the EVBMF-derived per-layer ranks for the ResNet-34 → ResNet-18 configuration on ImageNet-100. The EVBMF estimator processes 17 decomposable convolutional layers and produces a markedly non-uniform allocation. For CP, ranks span 30 (layer 5, the last 64-channel layer) to 218 (layer 17, the final 512-channel layer) in absolute terms. As a fraction of the output channel count, the allocation is highest in the early layers (59% for layer 1) and declines progressively to around 36– 44% in the deeper layers. It reflects the teacher ResNet-34's lower intrinsic rank utilisation in its wider, later stages relative to its early feature-extraction layers. A uniform 50% rank ratio would over-allocate parameters to the deeper redundant layers and under-allocate them to the early layers where the teacher concentrates its representational capacity.

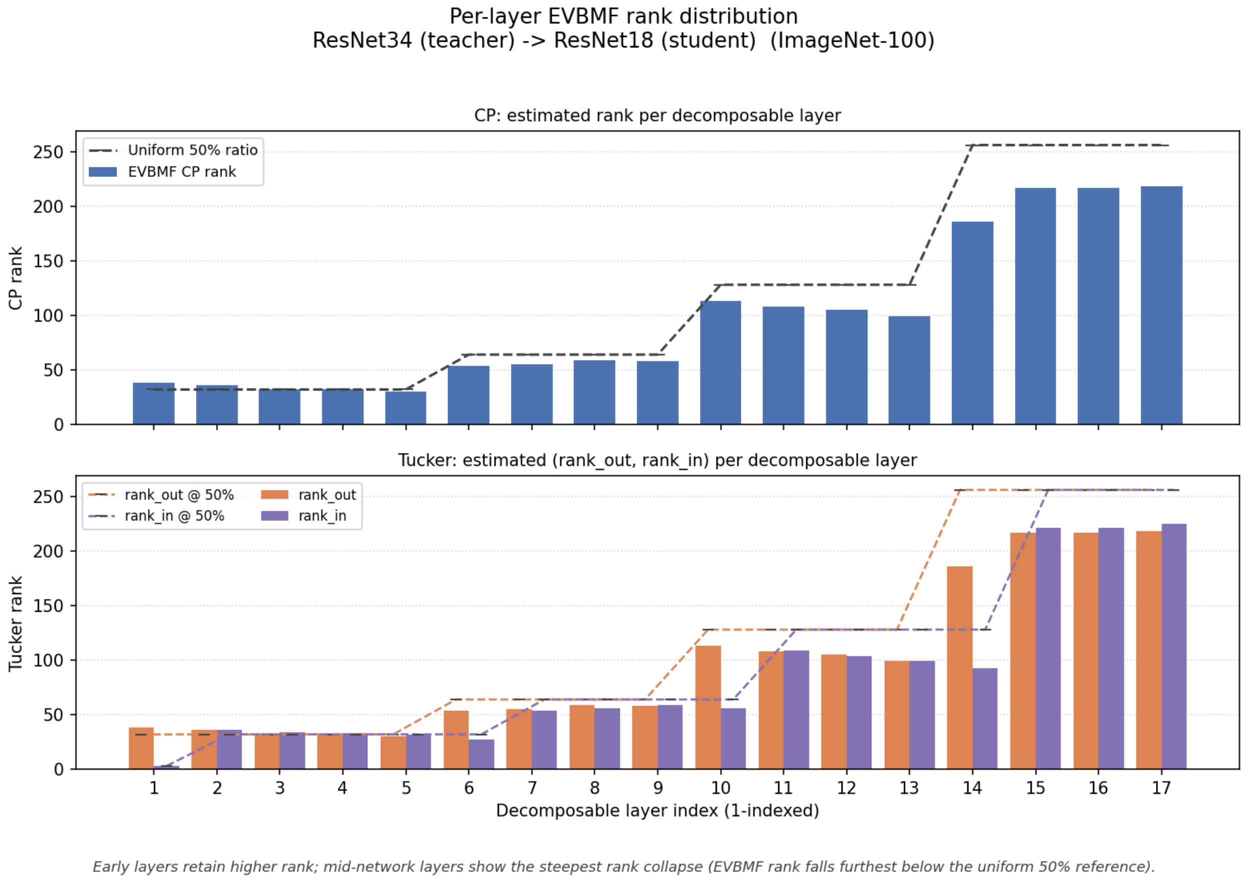

Figure 5: EVBMF-derived per-layer ranks for the CP and Tucker students (ResNet34 -> ResNet18, ImageNet-100). Tucker's output- and input-channel ranks are shown separately.

Tucker's (R_{out}, R_{in}) rank pairs reveal an additional structural pattern not visible in CP's single-rank curve; at the four architectural stage boundaries where the teacher doubles its channel count, layers 1 (conv1, 3→64), 6 (64→128), 10 (128→256) and 14 (256→512), the input-channel rank is substantially lower than the output-channel rank. Layer 14 for example, receives ranks $(186, 93)$: the teacher uses only 93 of 256 input dimensions effectively at this expansion point even though it produces 186 effective output dimensions. This asymmetry is estimated independently by mode-0 and mode-1 EVBMF unfoldings and transferred directly to the student. It gives Tucker per-stage control over input and output channel compression that CP's single rank cannot capture. The aggregate effect is the 6.56× compression ratio reported in Table 1 that is obtained without any manually tuned rank ratio.

5.6 Training Convergence

Figures 6 and 7 present the training convergence for the ResNet-34 → ResNet-18 configuration on ImageNet-100. Figure 6 shows all five distillation objectives for the Tucker student; Figure 7 isolates the AT objective and overlays both the CP and Tucker trajectories to allow direct comparison. The Tucker student leads the CP student at every epoch throughout the 100-epoch schedule. There is no crossover point at which CP overtakes Tucker. At epoch 5 (end of the LR warmup) Tucker already reaches 47.72% Top-1 while CP reaches only 22.62%, a gap of 25.1 percentage points. This early divergence reflects a capacity difference; Tucker's $3.25\ M$ parameters provide nearly 2.4× the parameter budget of CP's $1.33\ M$, allowing it to absorb the teacher's feature-level attention signals more rapidly during the high-learning-rate phase. The gap narrows over the main training phase (epochs 5–29) as CP's representations mature. It reduces to roughly 3 pp by epoch 29 (Tucker best 69.32% vs CP best 66.06%).

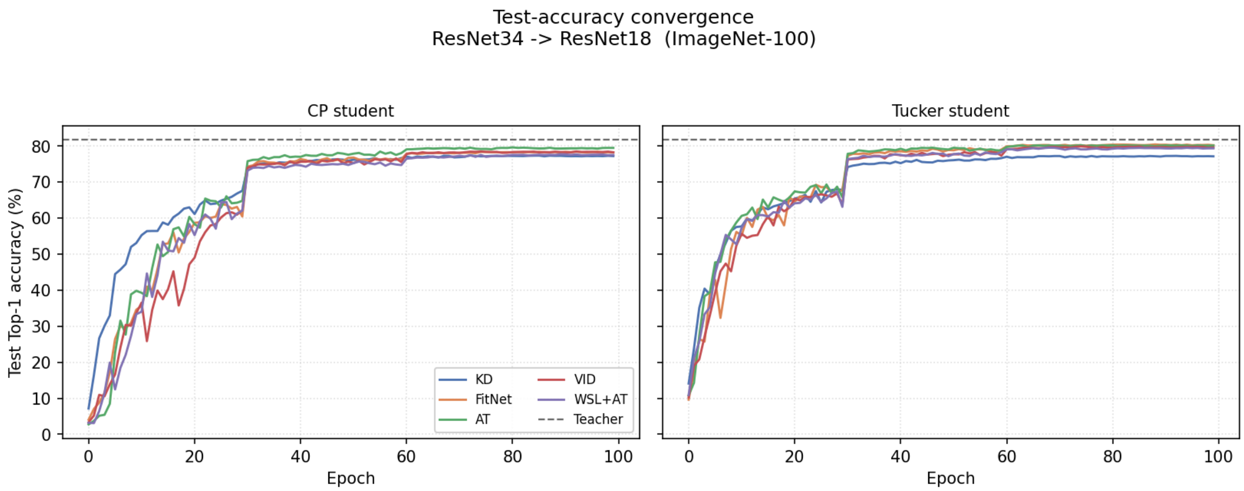

Figure 6: Validation Top-1 vs. epoch on ImageNet-100 (ResNet-34→ResNet-18) for five distillation objectives, with CP (left) and Tucker (right) students.

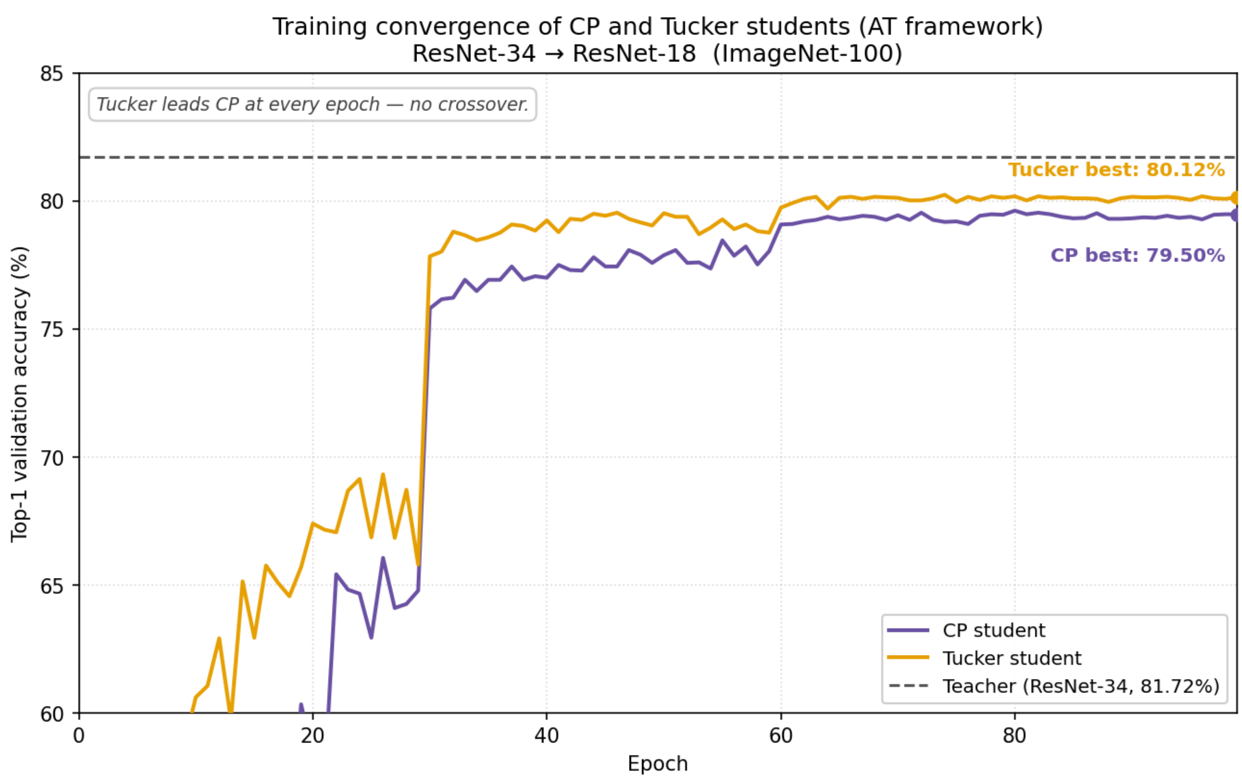

Figure 7: Validation Top-1 vs. epoch for the AT framework's CP (purple) and Tucker (amber) ResNet-18 students on ImageNet-100.

The first learning-rate decay at epoch 30 produces a step jump in both trajectories. CP rises from 64.78% to 75.80% and Tucker from 65.80% to 77.84% in a single epoch as the order-of-magnitude step-size reduction allows the optimiser to settle into a sharper minimum. After this decay Tucker maintains a consistent lead of 1.5–2.5 pp through epochs 30–59. The second decay at epoch 60 compresses the gap further; by epoch 70 Tucker reaches 80.12% while CP reaches 79.44% (0.68 pp). Both plateau after epoch 60; subsequent decays at epochs 80 and 90 produce only marginal gains. The final best-checkpoint accuracies are 80.24% for Tucker and 79.62% for CP, a gap of 0.62 pp that precisely matches the difference reported in Table 1 under the AT objective. Across all five distillation objectives in Figure 6, feature-based methods (FitNet, AT, VID) converge above KD and WSL. It shows that the value of intermediate-layer supervision irrespective of the specific loss formulation.

5.7 Automatic Loss Balancing

Figure 8 tracks the three effective task weights $e^{-s_{CE}}$, $e^{-s_{KL}}$, and $e^{-s_{AT}}$ as they evolve over 100 epochs under the homoscedastic uncertainty weighting of Section 4.7 for the CP student on ResNet-34 → ResNet-18. All three weights rise monotonically from their initialisation but at very different rates and to very different magnitudes. By epoch 99 the CE weight reaches 1.55, the KL weight 0.68 and the AT weight 702.7; a ratio of approximately $453{:}1{:}0.44$ (AT:CE:KL), meaning the mechanism applies a 453× larger multiplier to the attention-transfer term than to the cross-entropy term. The Tucker student reaches similar final values: CE 1.75, KL 0.76, AT 809.2 (ratio $463{:}1{:}0.43$). Even at epoch 0, before any meaningful learning, the AT weight already stands at 140 (CP) and 164 (Tucker). The mechanism recognises the scale imbalance immediately from the first gradient signal.

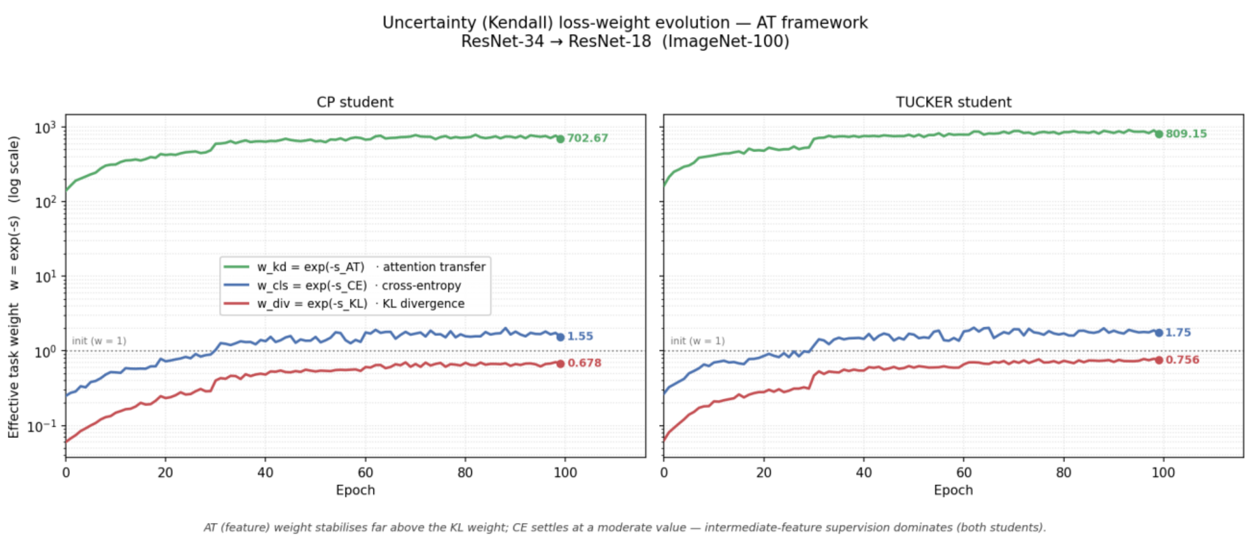

Figure 8: Evolution of the three effective task weights exp(-s) under uncertainty weighting (ResNet34 -> ResNet18, AT).

This large AT weight is not a signal that attention transfer is the dominant objective; it is a scale normalisation. The AT loss is the L2 distance between two unit-norm attention maps. At epoch 99 it measures 0.0013 per step, whereas the CE loss is 0.58 and the KL loss is 1.47. The mechanism multiplies the AT loss by 702.7 so that all three terms contribute comparably to the gradient. Concretely, the weighted contributions at epoch 99 are CE: $1.55 \times 0.58 = 0.90$, KL: $0.68 \times 1.47 = 1.00$, AT: $702.7 \times 0.0013 = 0.91$. They are within 10% of each other. The Kendall weighting therefore achieves its design goal. It converts three losses operating at wildly different numerical scales into three approximately equal gradient contributions and eliminates the need for a manual coefficient search. The fact that the weights converge to stable values rather than diverging confirms the effectiveness of the separate, weight-decay-free parameter group described in Section 4.7.

6 CONCLUSIONS AND FUTURE WORKS

6.1 Conclusions

This project set out to bridge the gap between the accuracy of modern deep convolutional networks and the limited compute, memory and energy budgets of edge-deployment platforms. The observation motivating the work is that the two dominant compression strategies (low-rank tensor decomposition and knowledge distillation) each fail in isolation. Sequential decompose-then-fine-tune pipelines suffer vanishing gradients and catastrophic accuracy loss at high compression ratios while logit-only distillation collapses under the capacity gap and cannot accommodate the dimensional mismatches that decomposition introduces. The framework developed here resolves both failure modes by unifying compression and distillation into a single joint training process. It constructs CP and Tucker decomposed students from scratch using per-layer ranks derived from the teacher's own weight spectrum via EVBMF, selects non-redundant supervision points through PURSUhInT representational clustering and co-trains both students under a shared frozen teacher with spatial attention transfer, output-space distillation and homoscedastic uncertainty weighting which requires neither trainable adapter modules nor manually tuned rank ratios or loss coefficients.

Evaluated across eight teacher-student configurations spanning residual, wide-residual and plain-convolutional architectures on CIFAR-100 and ImageNet-100, the framework consistently delivers order-of-magnitude compression at competitive accuracy. On ImageNet-100, the Tucker-decomposed ResNet-18 student reaches 80.12% Top-1 (within 1.60 percentage points of the ResNet-34 teacher) at 6.56× parameter compression while the CP student attains 16.04× compression at 79.50%. On CIFAR-100, compression ratios reach as high as 64.04× (ResNet-32×4 → WRN-16-2) while still retaining 72.01% Top-1. These reductions are realised in practice; single-image CPU inference latency for the ResNet-34 → ResNet-18 pair falls by 2.11× under Tucker decomposition.

Several consistent patterns emerged from the evaluation. The CP and Tucker decompositions form a reliable trade-off; CP achieves the higher compression ratio and Tucker the higher accuracy in every configuration with the ordering never reversing. Yet Tucker, despite its larger parameter count, achieves the lower inference latency owing to its shorter and denser operation sequence. Among the five distillation objectives evaluated within the identical pipeline, attention transfer proved the strongest overall and achieved the best Top-1 accuracy in 10 of the 16 evaluated student columns (7 of 12 on CIFAR-100 and 3 of 4 on ImageNet-100). Feature-based supervision outperformed logit-only distillation in nearly every configuration and it shows that intermediate-layer guidance is essential for training heavily compressed students.

The two automatic components behaved as designed. The EVBMF estimator produced markedly non-uniform, teacher-calibrated rank allocations that a single global ratio could not reproduce and the uncertainty weighting converged to effective weights that despite spanning three orders of magnitude, equalised the three losses' gradient contributions to within ten percent of one another while eliminating manual coefficient search entirely. Taken together, these results demonstrate that jointly optimising structural compression and knowledge transfer is a viable and robust alternative to the fragile sequential pipeline, capable of producing deployable, accurate and substantially compressed networks across diverse architectures without architecture-specific tuning.

6.2 Future Work

While the framework achieves its primary objectives, several directions remain open for extending and more fully characterising it.

The present study adopts PURSUhInT hint points throughout but does not isolate their contribution. A controlled comparison against depth-uniform, final-block and random layer-selection baselines (measuring both convergence rate and final accuracy) would quantify how much of the framework's performance is attributable to non-redundant hint selection specifically, as originally envisaged in the evaluation methodology of Section 3.4.

The reported results are main runs only. A full ablation over the number of hint points K, the EVBMF rank-scaling factor, the individual loss components (cross-entropy, KL divergence and attention transfer) and fixed versus uncertainty-based loss weighting would clarify the framework's sensitivity to each design choice and identify which components contribute most to the observed gains.

Evaluation was conducted on ImageNet-100 which a 100-class subset. Extending the framework to the full ImageNet-1k benchmark would test whether the compression-accuracy trade-off holds under greater task difficulty and class granularity where the capacity gap between the compressed student and the teacher is more severe.

The framework currently targets convolutional networks. Adapting the decomposition and attention-transfer components to vision transformers, for instance by factorising the projection and feed-forward weight matrices and aligning attention distributions across layers, would broaden its applicability to the architectures now dominant in vision and multimodal tasks.

Tensor decomposition is orthogonal to quantisation and pruning. Combining the present framework with post-training quantisation or structured pruning could compound the compression ratios achieved here and push students toward the still tighter budgets of microcontroller-class hardware.

Latency was measured on a single CPU core as a proxy for the edge scenario. Deploying the compressed students on representative target hardware (mobile systems-on-chip, embedded GPUs or microcontrollers) and profiling real energy consumption and memory bandwidth would validate the practical deployment benefits beyond parameter and FLOP counts.

Attention transfer was the strongest objective overall but not universally dominant. Logit-only distillation prevailed on the VGG-13 → VGG-8 pair, and other objectives led at the highest compression ratios. A mechanism that selects or combines distillation objectives adaptively per configuration rather than fixing a single objective, could further improve robustness across architecture families.

7 REFERENCES

[1] A. Gholami, Z. Yao, S. Kim, C. Hooper, M. W. Mahoney, and K. Keutzer, "AI and Memory Wall," *IEEE Micro*, vol. 44, no. 3, pp. 33–39, May 2024, doi: 10.1109/MM.2024.3373763.

[2] V. Lebedev, Y. Ganin, M. Rakhuba, I. Oseledets, and V. Lempitsky, "Speeding-up Convolutional Neural Networks Using Fine-tuned CP-Decomposition," arXiv.org. Accessed: Jun. 04, 2026. [Online]. Available: https://arxiv.org/abs/1412.6553v3

[3] Y.-D. Kim, E. Park, S. Yoo, T. Choi, L. Yang, and D. Shin, "Compression of Deep Convolutional Neural Networks for Fast and Low Power Mobile Applications," arXiv.org. Accessed: Jun. 04, 2026. [Online]. Available: https://arxiv.org/abs/1511.06530v2

[4] J. Gusak *et al.*, "Automated Multi-Stage Compression of Neural Networks," presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops, 2019, pp. 0–0. Accessed: Jun. 04, 2026. [Online]. Available: https://openaccess.thecvf.com/content_ICCVW_2019/html/LPCV/Gusak_Automated_Multi-Stage_Compression_of_Neural_Networks_ICCVW_2019_paper.html

[5] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," arXiv.org. Accessed: Jun. 04, 2026. [Online]. Available: https://arxiv.org/abs/1503.02531v1

[6] S. I. Mirzadeh, M. Farajtabar, A. Li, N. Levine, A. Matsukawa, and H. Ghasemzadeh, "Improved Knowledge Distillation via Teacher Assistant," *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 04, pp. 5191–5198, Apr. 2020, doi: 10.1609/aaai.v34i04.5963.

[7] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, "FitNets: Hints for Thin Deep Nets," Mar. 27, 2015, *arXiv*: arXiv:1412.6550. doi: 10.48550/arXiv.1412.6550.

[8] R. K. Keser, A. Ayanzadeh, O. A. Aghdam, C. Kilcioglu, B. U. Toreyin, and N. K. Ure, "PURSUhInT: In Search of Informative Hint Points Based on Layer Clustering for Knowledge Distillation," *Expert Syst. Appl.*, vol. 213, p. 119040, Mar. 2023, doi: 10.1016/j.eswa.2022.119040.

[9] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton, "Similarity of Neural Network Representations Revisited," in *Proceedings of the 36th International Conference on Machine Learning*, PMLR, May 2019, pp. 3519–3529. Accessed: Jun. 04, 2026. [Online]. Available: https://proceedings.mlr.press/v97/kornblith19a.html

[10] S. Nakajima, M. Sugiyama, C. T. A. Jp, S. D. Babacan, and R. Tomioka, "Global Analytic Solution of Fully-observed Variational Bayesian Matrix Factorization".

[11] S. Zagoruyko and N. Komodakis, "Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer," arXiv.org. Accessed: Jun. 04, 2026. [Online]. Available: https://arxiv.org/abs/1612.03928v3

[12] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," 2009.

[13] O. Russakovsky *et al.*, "Imagenet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, 2015.

[14] S. Han, H. Mao, and W. J. Dally, "Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding," arXiv.org. Accessed: Jun. 04, 2026. [Online]. Available: https://arxiv.org/abs/1510.00149v5

[15] B. Jacob *et al.*, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 2704–2713.

[16] J. Guo, M. Chen, Y. Hu, C. Zhu, X. He, and D. Cai, “Reducing the teacher-student gap via spherical knowledge disitllation,” *ArXiv Prepr. ArXiv201007485*, 2020.

[17] B. Zhao, Q. Cui, R. Song, Y. Qiu, and J. Liang, “Decoupled knowledge distillation,” in *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition*, 2022, pp. 11953–11962.

[18] H. Zhou *et al.*, “Rethinking soft labels for knowledge distillation: A bias-variance tradeoff perspective,” *ArXiv Prepr. ArXiv210200650*, 2021.

[19] S. Ahn, S. X. Hu, A. Damianou, N. D. Lawrence, and Z. Dai, “Variational information distillation for knowledge transfer,” presented at the Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019, pp. 9163–9171.

[20] M. Ji, B. Heo, and S. Park, “Show, attend and distill: Knowledge distillation via attention-based feature matching,” presented at the Proceedings of the AAAI conference on artificial intelligence, 2021, pp. 7945–7952.

[21] C. J. Hillar and L.-H. Lim, “Most tensor problems are NP-hard,” *J. ACM JACM*, vol. 60, no. 6, pp. 1–39, 2013.

[22] A. Kendall, Y. Gal, and R. Cipolla, “Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7482–7491.